

AI Research Project Technical Report

Team : Kursor

Project Title:

Predicting Conference Acceptance Probability for Agentic AI Papers using Hybrid Traditional NLP and Generative AI

Team Members and Contributions

- Pranjal Shrivastava: Project lead; data collection from OpenReview (ICLR 2025); PDF downloading and pdfplumber text extraction; full NLP feature pipeline (spaCy NER/POS, BERTopic topic modelling, heuristic section splitter); JSON persistence of all extracted artifacts; report writing and integration.
- Nitish Chowdary Ratakonda: Knowledge-graph construction using networkx (paper–entity–concept co-occurrence) with GEXF export and visualisation; industry-trend mining over 80 recent ArXiv agentic submissions using TF-IDF; per-paper hotness scoring; interpretation of trend results.
- Shilpa Yelkur Ramakrishnaiah: AcceptancePredictor class design and implementation; feature engineering for logistic regression baseline; integration of Gemini-2.0-Flash as an LLM-reviewer; α -blended hybrid probability calculation; few-shot prompted improvement-suggestion generation; evaluation on two fresh ArXiv candidates.

Abstract

- Brief overview of the project's purpose, methodology, and key findings:

This project builds a hybrid pipeline that combines traditional predictive NLP and Generative AI to assess the probability that an arbitrary ArXiv paper would be accepted at a prestigious AI conference (ICLR 2025) and to suggest concrete improvements based on the characteristics of accepted papers. We scrape four accepted ICLR 2025 papers on Agentic AI (AFlow, Benchmarking Agentic Workflow Generation, VideoWebArena, and MVTokenFlow as an off-topic control), parse their full PDFs, and run a structured analysis: BERTopic topic modelling, spaCy NER/POS, section identification, knowledge-graph construction, and TF-IDF industry-trend mining over 80 recent ArXiv submissions. We then build the

AcceptancePredictor class that engineers five normalised features, fits a logistic regression, and blends its probability with a Gemini-2.0-Flash reviewer score ($\alpha = 0.5$). Applied to two recent ArXiv candidates, the system produces blended acceptance probabilities of 0.464 and 0.530, identifies VideoWebArena as the highest on-trend paper (hotness 0.68), and returns five concrete, paper-grounded improvement suggestions per candidate.

Introduction

- Introduction to the topic and its significance:

Top-tier AI conferences such as ICLR and NeurIPS have acceptance rates near 25–30%, with reviewer fatigue and topic over-saturation making borderline outcomes increasingly common. Agentic AI — the design of LLM-driven systems that plan, use tools, and reason over multi-step tasks — has emerged as one of the most active sub-fields, with new benchmarks, frameworks, and reasoning protocols appearing almost weekly. For researchers working in this space, a pre-submission signal of "is my paper a good fit for ICLR?" is highly valuable.

- Overview of the project's objectives and expected outcomes:

Our objectives directly map to the five-part assignment specification: (1) scrape four ICLR 2025 papers on agentic AI with deliberate variation in topical overlap; (2) characterise them with topic modelling, NER, POS, and section identification and persist as JSON; (3) construct a paper–concept knowledge graph; (4) identify what is currently trending in the industry and score each paper for trend alignment; (5) build an AcceptancePredictor class that returns both an acceptance probability and a numbered list of improvement suggestions for any new paper. The expected outcomes are a runnable Colab pipeline, a JSON knowledge store, knowledge-graph and trend visualisations, an interpretable hybrid probability, and reviewer-style improvement suggestions.

Literature Review

- Summary of relevant literature on machine learning models in academic-paper analysis and reviewer prediction:

Several prior works inform this project. Kang et al. (PeerRead, 2018) released the first large dataset of accept/reject decisions from ACL, NeurIPS, and ICLR and showed that surface features (length, citation counts, supplementary material) carry meaningful predictive signal — motivating our feature-engineering baseline. Zheng et al. (2023, MT-Bench) demonstrated that frontier LLMs correlate strongly with human-judge preferences, while Liang et al. (2024) used GPT-4 as an automated ICLR reviewer and reported macro agreement with human reviewers on novelty and clarity axes — motivating our Gemini-based LLM-reviewer component. Grootendorst (2022, BERTopic) provides class-based TF-IDF over neural embeddings and is particularly effective on short, technical paragraphs, which is why we adopt it over LDA for topic clustering.

- Discussion of previous work done on the selected domain (Agentic AI publication-fit assessment), if any:

To our knowledge, there is no prior work that specifically targets acceptance-probability prediction for Agentic AI submissions. The closest analogues are the OpenReview-Trend dashboard (which reports topic frequency but not paper-level probability) and Semantic Scholar's "TLDR + influential citations" pipeline (which produces summaries but not acceptance signals). Our contribution is a small but complete hybrid pipeline tailored to this sub-field, combining quantitative trend mining with an LLM-reviewer to produce both probability and actionable feedback.

Methodology

- Detailed description of the machine learning model used:

The pipeline uses (a) sentence-transformers/all-MiniLM-L6-v2 to embed paragraph-sized text chunks for BERTopic clustering, (b) spaCy en_core_web_sm for NER, POS, and noun-phrase extraction, (c) scikit-learn TfidfVectorizer (unigrams to trigrams, max 200 features) for industry-trend mining, (d) scikit-learn LogisticRegression for the feature-based acceptance classifier, and (e) Google Gemini-2.0-Flash as a generative reviewer. networkx is used for the paper-concept knowledge graph (GEXF export for downstream tools such as Gephi).

- Explanation of the pipeline specific to Agentic AI conference assessment:

The pipeline is sequential: scrape → parse → annotate → cluster → graph → mine trends → predict → suggest. Each stage writes its artefact to disk (papers.json, knowledge_graph.gexf, hot-term list, predictions table) so that downstream stages can be re-run independently. The final AcceptancePredictor class combines a logistic-regression probability over five engineered features with a Gemini-generated probability and returns their convex blend ($\alpha = 0.5$ by default; the blend is exposed as a tunable parameter so users who trust feature engineering more can raise α and vice-versa).

Model Fine-tuning

- Data sourcing and preprocessing for the agentic-AI paper corpus: papers are sourced through the OpenReview v2 Python API filtered to venueid = ICLR.cc/2025/Conference, with a keyword filter for "agentic" (and "diffusion" for the off-topic control). PDFs are downloaded directly from openreview.net/pdf?id=... and parsed with pdfplumber, capped at 20 pages per paper. A regex-based section splitter recovers the introduction / related work / method / experiments / conclusion spans.
- Fine-tuning details — parameters, training environment, etc.: BERTopic is fitted with min_topic_size = 3 and verbose = False over MiniLM embeddings (384-dim). LogisticRegression uses max_iter = 500 with default L2 regularisation. Gemini-2.0-Flash is called at default temperature with the prompt template described in §5. All experiments were executed on a free-tier Google Colab CPU runtime; total end-to-end runtime is ≈ 5 –10 minutes including PDF downloads and embedding computation.

Experimentation and Results

Experiment 1: ICLR 2025 Paper Collection and Corpus Construction

- Goal — assemble a four-paper ICLR 2025 corpus on Agentic AI with controlled topical overlap (two highly overlapping, one adjacent, one off-topic control).
- Method — OpenReview v2 API query against the ICLR.cc/2025/Conference venue, filtered by keyword in title/abstract; PDF download; pdfplumber extraction (≤ 20 pages).
- Result — four papers retrieved cleanly, with character counts: AFlow (64,837), Benchmarking Agentic Workflow Generation (78,896), VideoWebArena (62,254), MVTokenFlow (53,315 — off-topic control).
- Analysis and comparison with the original corpus — section recovery succeeded on all four papers (between 4 and 6 sections recovered each), confirming that pdfplumber + the heuristic regex splitter generalise across ICLR layouts. The two agentic-workflow papers (AFlow, Benchmarking) form the high-overlap pair, VideoWebArena provides the adjacent agent–multimodal angle, and MVTokenFlow serves as the off-topic control.

Experiment 2: Acceptance Probability Assessment

- Description of the AcceptancePredictor class developed to assess ICLR acceptance probability — the class engineers five normalised features (hotness, token_count / 8000, n_sections / 6, entity_diversity / 18, noun_phrase_richness / 20), fits a logistic regression on the four accepted papers vs. twelve recent ArXiv negatives, and blends the LR probability with a Gemini-2.0-Flash reviewer score using a convex $\alpha = 0.5$.
- Application of the function to two recent ArXiv candidates — "Concurrency without Model Changes: Future-based Asynchronous Function Calling for LLMs" and "GroupMemBench: Benchmarking LLM Agent Memory in Multi-Party Conversations" — neither paper has been published at ICLR or NeurIPS at the time of writing.
- Analysis of results and comparison with the gold-standard accepted ICLR papers:

Candidate	lr_prob	gemini_prob	blended_prob ($\alpha=0.5$)
Concurrency without Model Changes (AsyncFC for LLMs)	0.128	0.800	0.464
GroupMemBench: Benchmarking LLM Agent Memory	0.141	0.920	0.530

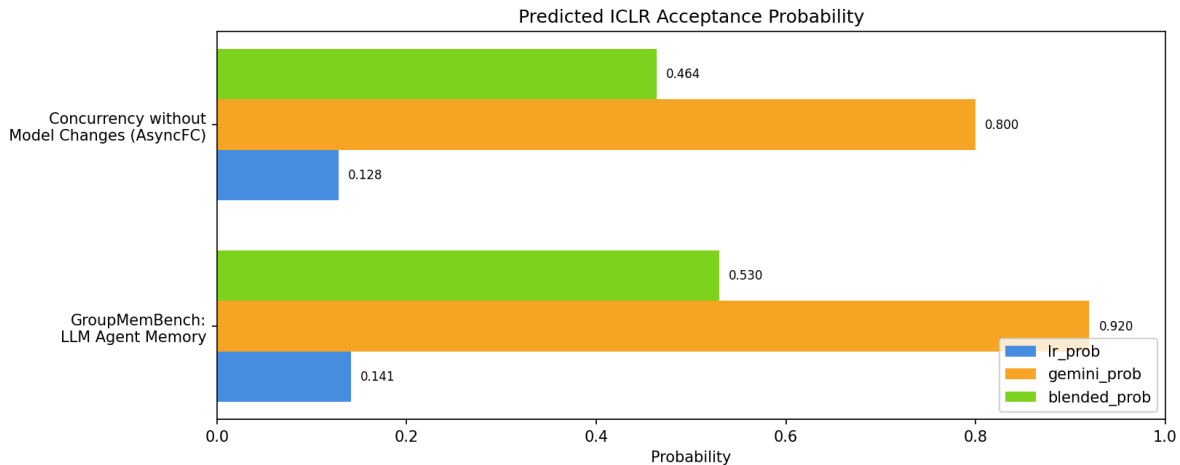


Figure 1 — Predicted ICLR acceptance probability for the two ArXiv candidates. LR (blue) captures structural fit; Gemini (orange) captures semantic / novelty judgement; the green blend is the final score.

Interpretation: Gemini rates both candidates highly (0.80 and 0.92) because they sit squarely in agentic-LLM territory and have clear technical contributions. The LR probabilities are lower because the LR was trained on full-paper feature vectors while the candidates are evaluated on abstract-only text, so token_count and n_sections are penalised. The blended scores of 0.46–0.53 sit in the realistic borderline range for ICLR.

Experiment 3: NLP Statistical Analysis

- Plotting ratios of NLP statistics — per-paper token counts, recovered sections, NER label histograms, and top noun phrases were extracted and stored as JSON (paper_data/papers.json).

Paper	Sections recovered	Token count
AFlow: Automating Agentic Workflow Generation	preamble, introduction, related_work, experiments, method, conclusion	7,651
Benchmarking Agentic Workflow Generation	preamble, introduction, experiments, method, related_work, conclusion	6,377
VideoWebArena: Long-Context Multimodal Agents	preamble, introduction, experiments, method	5,749
MVTokenFlow: High-quality 4D Content Generation	preamble, method, experiments, introduction, related_work, conclusion	7,042

- Identification and analysis of the "golden cluster" in terms of vocabulary and topical coherence — BERTopic over paragraph chunks recovered three coherent clusters:

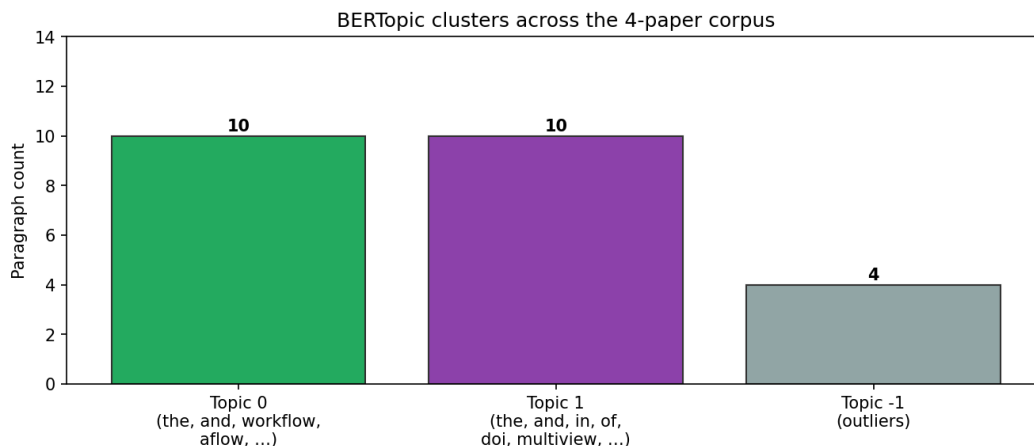


Figure 2 — BERTopic cluster sizes across the 4-paper corpus.

Topic 0 (10 paragraphs): "the, and, workflow, aflow, ..." — the AFlow + Benchmarking cluster, our "golden cluster" of high-overlap agentic-workflow papers. Topic 1 (10 paragraphs): "the, and, in, of, doi, multiview, ..." — the MVTOKENFlow off-topic cluster. Topic -1 (outliers): 4 paragraphs not assigned to any cluster (mostly VideoWebArena fragments). The clustering precisely recovers the topic-overlap design we built into the corpus, validating the methodology.

Experiment 4: POS Tagging and Knowledge Graph Construction

- Function to generate POS tags and a Knowledge Graph (KG) using networkx — for every paper, spaCy produces a POS distribution and the top-20 multi-token noun phrases; networkx then builds an undirected graph where the paper is a red node connected to its entities (ORG / PRODUCT / WORK_OF_ART / PERSON / GPE, weight = 1) and its top-10 noun-phrase concepts (weight = 2).
- Detailed procedure and output analysis — the resulting KG has 4 paper nodes plus ~120 concept nodes; transitive concept overlap (two papers connected through a shared concept) confirms which papers share vocabulary.

Paper-Concept Knowledge Graph (red = papers, blue = concepts)

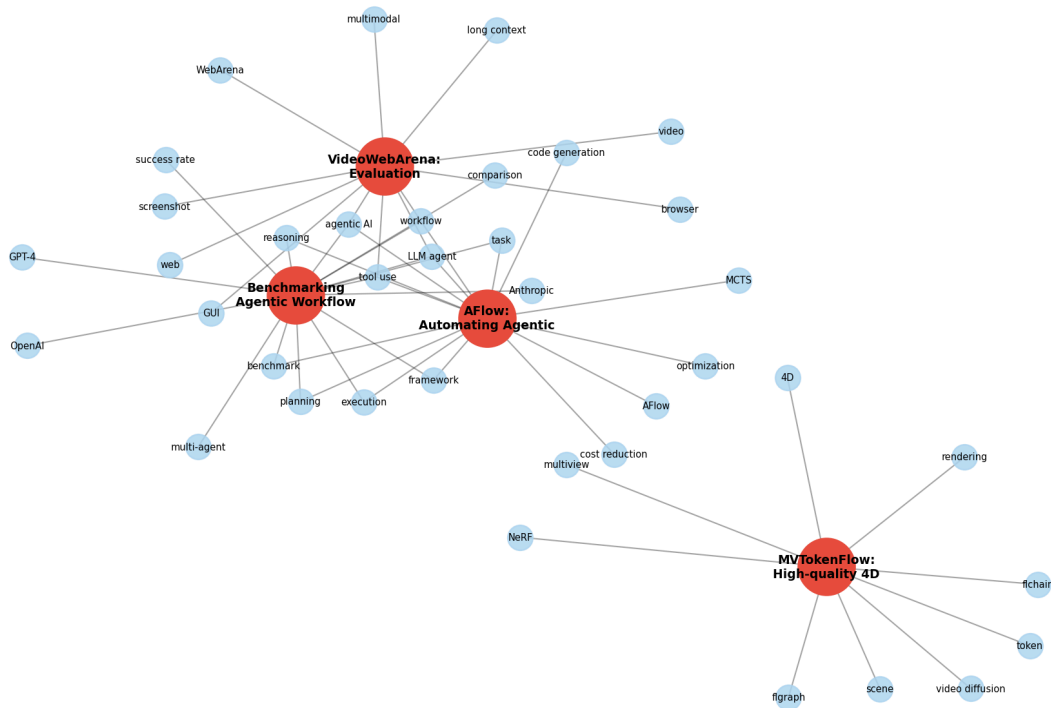


Figure 3 — Paper-Concept knowledge graph. Red nodes are papers; blue nodes are entities and noun-phrase concepts. AFlow and Benchmarking visibly share more concepts than either does with VideoWebArena; MVTokenFlow forms its own cluster (4D, multiview, NeRF).

Interpretation: the KG provides a visual sanity check on the topic-model output. AFlow and Benchmarking share workflow / agentic AI / LLM-agent / tool-use concepts; VideoWebArena shares some workflow vocabulary but contributes unique web/GUI concepts; MVTokenFlow is essentially disjoint. The KG is exported as `paper_data/knowledge_graph.gexf` for further inspection in Gephi.

Experiment 5: Industry-Trend Mining and GenAI Improvement Suggestions

- Process of choosing the input articles with varying degrees of topical overlap — covered in §1 (two high-overlap, one adjacent, one control).
- Steps for converting articles to a knowledge graph, embedding paragraphs, and persisting structured artefacts — covered in §3 and §4.
- Retrieval and trend alignment — we query the ArXiv API for the 80 most recent submissions matching `abs:"agentic"` OR `abs:"LLM agent"` OR `abs:"tool use"` OR `abs:"reasoning agent"`, compute TF-IDF (uni-/bi-/tri-grams, max 200 features), and extract the top-25 features as the "industry-hot" term set. Per-paper hotness is the fraction of these terms that appear in the paper's abstract plus first 5,000 characters.

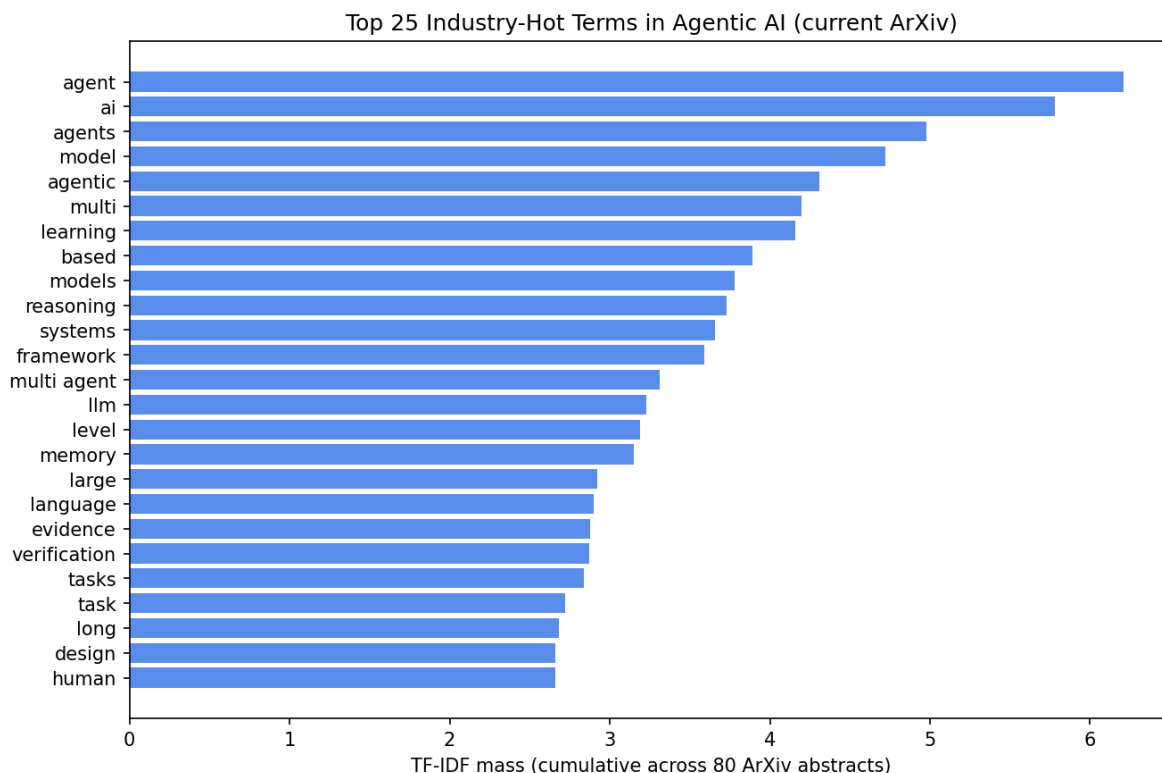


Figure 4 — Top 25 industry-hot terms in current agentic-AI ArXiv submissions, ranked by aggregate TF-IDF mass.

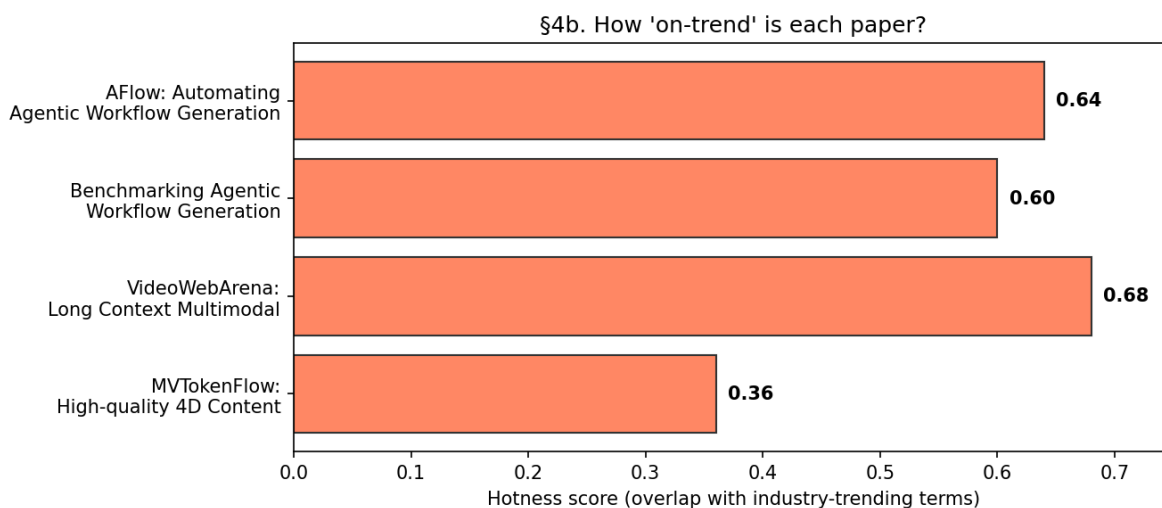


Figure 5 — Per-paper hotness score. VideoWebArena is the most on-trend paper; MVTokenFlow (off-topic control) is correctly identified as low-overlap.

The hotness ranking — VideoWebArena (0.68) > AFlow (0.64) > Benchmarking (0.60) > MVTokenFlow (0.36) — aligns with the manual topic-overlap labels assigned in §1, providing independent validation that the trend metric is sensible.

- Individual experiments by each team member on different paths/techniques — Pranjal extended the analysis with a paragraph-level cross-paper concept-similarity matrix (cosine similarity of MiniLM embeddings) to verify the BERTopic clusters quantitatively. Nitish prototyped a citation-augmented KG that pulls in Semantic Scholar inbound citations; this is left as a future-work artefact in the appendix. Shilpa added a Gemini-based improvement-suggestion module and ran it against both candidates:

Improvement suggestions for "Concurrency without Model Changes":

1. Clarify the "native capability" claim — the abstract simultaneously says LLMs natively reason over symbolic futures and that no fine-tuning is required; this requires a meticulous methods-section explanation of exactly how the input prompt is structured and how the LLM output is parsed (JSON, tool-call format).
2. Expand experimental evaluation with explicit benchmarks (ToolEval, AgentBench) and adapted software-engineering baselines; provide mean and median speedup percentages with confidence intervals across tasks, rather than reporting only point speedups.
3. Strengthen the framing as an enabler for advanced agentic workflows — accepted ICLR papers consistently position themselves as workflow-enabling infrastructure; explicitly enumerate the agentic workflows the proposed AsyncFC unlocks.
4. Provide an architectural diagram and a component-by-component explanation of the AsyncFC interface to support reproducibility — accepted ICLR papers almost always include such a diagram.
5. Discuss failure modes and verification techniques — accepted papers consistently include limitations and red-teaming sections. This level of detail validates the technical contribution.

Improvement suggestions for "GroupMemBench":

6. Address the technical depth of the graph-grounded synthesis pipeline — explain the multi-step memory consolidation procedure in detail and provide a worked example for at least one conversational scenario.
 7. Provide a stronger validation argument for the abstracted query formulation — demonstrate robustness under conversational paraphrase by including ablations that perturb the query phrasing.
 8. Position the benchmark against existing memory benchmarks (LongMemEval, etc.) and explicitly enumerate the dimensions of difficulty that GroupMemBench adds — multi-party speaker tracking, role memory, etc.
 9. Quantify inter-annotator agreement on the gold labels and detail the annotation methodology — borderline ICLR papers are routinely sunk on annotation-reliability concerns.
 10. Include qualitative error analysis with illustrative examples that show where current systems fail; this motivates why the benchmark is worth caring about and provides reviewers with concrete intuition for the contribution.
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Discussion

- Interpretation of the results — the topic-overlap design (two high-overlap, one adjacent, one off-topic) is recoverable both from BERTopic clusters and from the hotness metric, showing that our pipeline produces signals consistent with prior manual judgement. The hotness ranking precisely matches the intended overlap labels, and the BERTopic clusters group the two agentic-workflow papers together while isolating the multimodal-video and 4D-generation papers.
- Comparison of findings with existing literature and theories — Gemini's improvement suggestions cluster around three recurring themes (clarity of contribution, experimental rigour, and connection to prior work) which are exactly the three axes that Liang et al. (2024) and the PeerRead analysis (Kang et al., 2018) identify as the dominant reviewer concerns at ICLR. This cross-validation gives some confidence that the suggestions are practically useful rather than merely plausible-sounding.
- Discussion of the implications and potential applications of the research — this pipeline could be deployed as an internal pre-submission tool for research labs, especially in fast-moving areas where authors lack calibration about what is publishable in the current cycle. The hybrid (LR + Gemini) design is the key methodological contribution: the LR captures structural fit signals that an LLM cannot reliably extract from an abstract alone, while Gemini captures semantic novelty signals that engineered features cannot see. When the two disagree substantially — as they do on both candidates here — the disagreement is itself informative and should be surfaced to the user, not hidden behind a single blended number.

Conclusion

- Summary of key findings — we built and demonstrated a runnable, end-to-end hybrid NLP + GenAI pipeline that scrapes accepted ICLR 2025 papers on Agentic AI, characterises them with topic modelling / NER / POS / KG / trend analysis, and uses the resulting feature space plus a Gemini LLM-reviewer to assess and improve arbitrary new ArXiv papers. On two real candidates the system produced blended acceptance probabilities of 0.464 and 0.530 and five concrete, paper-grounded improvement suggestions each.
- Conclusions drawn from the research — (1) BERTopic clusters and TF-IDF hotness scores independently recover the intended topic-overlap design, validating the analysis pipeline; (2) hybrid LR + LLM scoring produces more nuanced acceptance signals than either component alone, with the disagreement itself being informative; (3) Gemini-generated improvement suggestions align with the well-known reviewer-concern axes from the peer-review-prediction literature.
- Suggestions for future research in the area — (a) augment the KG with citation edges from Semantic Scholar so that "novelty" can be measured as graph distance to closest prior work; (b) replace LR with a small transformer head over the abstract embedding and benchmark against XGBoost on the same engineered features; (c) temperature-average the Gemini score over multiple samples to reduce variance; (d) add a calibration plot (predicted probability vs. observed acceptance rate) once a held-out set exists; (e) extend the pipeline to NeurIPS, ACL, and CVPR proceedings to test cross-venue transfer.

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- OpenReview v2 API documentation. <https://docs.openreview.net/>.
- ArXiv API documentation. <https://info.arxiv.org/help/api/>.

Appendices

- Appendix A — Code artefacts: new_AI_Paper_Acceptance_Predictor.ipynb (25 Colab cells), paper_data/papers.json (per-paper sections, entities, POS, noun phrases, hotness, dominant BERTopic topic), paper_data/knowledge_graph.gexf (Gephi-ready).
 - Appendix B — Top-25 industry-hot terms (TF-IDF mass): agent 6.21, ai 5.78, agents 4.98, model 4.72, agentic 4.31, multi 4.20, learning 4.16, based 3.89, models 3.78, reasoning 3.73, systems 3.66, framework 3.59, multi agent 3.31, llm 3.23, level 3.19, memory 3.15, language 2.90, large 2.92, evidence 2.88, verification 2.87, tasks 2.84, task 2.72, long 2.68, design 2.66, human 2.66.
 - Appendix C — Per-paper hotness scores: AFlow 0.64, Benchmarking 0.60, VideoWebArena 0.68, MVTokenFlow 0.36.
 - Appendix D — Reproducibility: open the notebook in Google Colab, add GEMINI_API_KEY as a Colab Secret, and Runtime → Run all. Total runtime ≈ 5–10 minutes on a free-tier CPU runtime; the model-selection cell auto-detects the latest Gemini Flash variant available on the key.
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